

## RESEARCH ARTICLE SUMMARY

## BIOCHEMISTRY

# Landscape profiling of PET depolymerases using a natural sequence cluster framework

Hogyun Seo<sup>†</sup>, Hwaseok Hong<sup>†</sup>, Jiyoung Park<sup>†</sup>, Seul Hoo Lee, Dongwoo Ki, Aejin Ryu, Hye-Young Sagong, Kyung-Jin Kim\*

**INTRODUCTION:** Numerous PET-degrading enzymes (PETases) have been discovered in nature, including PETases from *Ideonella sakaiensis* (IsPETase) and leaf-compost cutinase (LCC), which are undergoing engineering for industrial applications. However, little is known about how many enzymes possess this capability or which enzymes have greater potential for degrading plastics. The natural diversity accessible through sequence databases may hold the key to addressing this issue, but so far there has been a lack of available approaches for identifying the fitness of promiscuous enzymes from sequence data.

**RATIONALE:** We have attempted to determine the distribution of fitness of a large set of enzymes, providing a comparative and holistic view of the evolutionary success of naturally occurring PETases. To organize the landscape of the polyesterase-lipase-cutinase family, we generated a two-dimensional semantic network with sequence homology distances analyzed by a neighborhood analysis module, which controls the network stringency by using distance histogram data for each sequence. On the basis of the connected

components (clusters) and density distribution of nodes on the network, representative nodes were sampled with stratified and cluster sampling methods to experimentally measure fitness for PET-degrading activity and thermal stability.

**RESULTS:** The landscaping approach generated 170 clusters comprising 1894 nodes (2064 sequences) of the polyesterase-lipase-cutinase family, covering the majority of known PETases. The primary screening for PETase, using stratified sampling, showed that more than half of the selected nodes have detectable PET-degrading activity, with a few nodes possessing high fitness. The secondary screening for PETase, using cluster sampling, distinguished three clusters (C3, C25, and C158) containing nodes with high fitness, demonstrating the representativeness of the selected nodes by their positions. Out of the 158 nodes that participated in the screening rounds, N1319 (Mipa-P) from C158 and N1501 (Kubu-P) from C25 were identified as the most promising enzymes, exhibiting superior features in PET decomposition speed and durability compared with other known benchmarks. Further engi-

neering with rationally selected cross-template mutations generated the Mipa-P<sup>M19</sup> and Kubu-P<sup>M12</sup> variants. The engineered Kubu-P<sup>M12</sup> outperformed the engineered benchmarks under high substrate loads in industrial conditions. Kubu-P<sup>M12</sup> even endured ethylene glycol as the solvent and reactant in enzyme-catalyzed PET glycolysis, yielding bis(2-hydroxyethyl) terephthalate as the main product at concentrations reaching tens of millimolar.

**CONCLUSION:** The landscaping-and-sampling framework developed in this study provides an effective method for exploring the fitness landscape of promiscuous protein families, facilitating the accumulation of sequence-to-activity data. This study identifies 90 previously unreported sequences with potential PETase activity, of which approximately 90% represent novel groups. These findings provide a broad overview of the distribution of the PETases within the constructed library, enabling relative fitness comparisons. Consequently, this analysis led to the identification of three unexplored protein lineages exhibiting high fitness for PETase activity and to identification of the most potent proteins, Mipa-P and Kubu-P. Notably, the Kubu-P<sup>M12</sup> variant engineered from a high potential template outperforms benchmarks under industrially relevant conditions. ■

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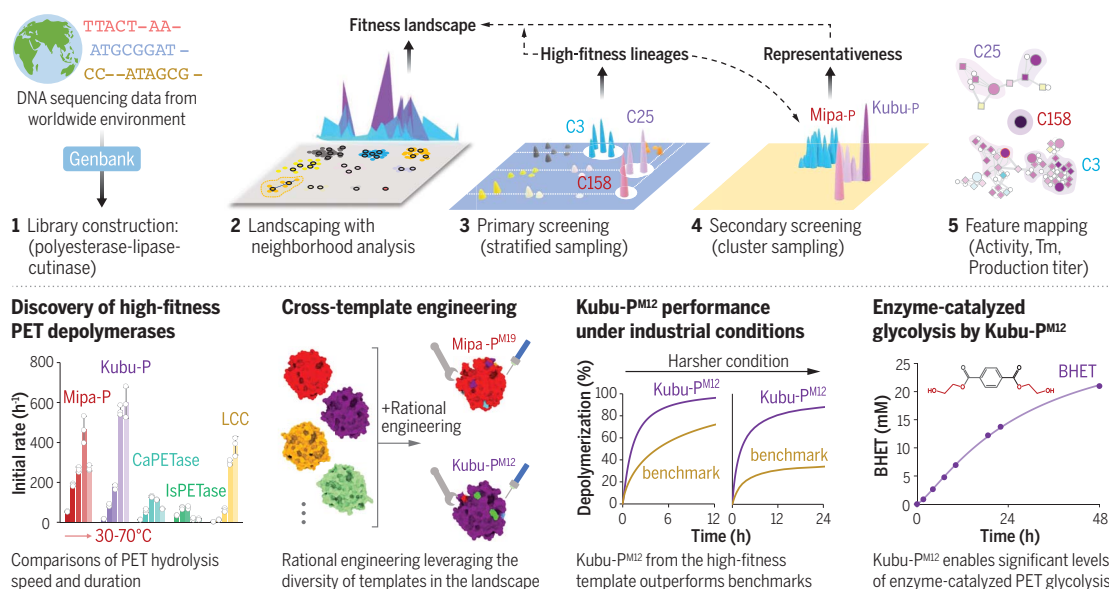
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**Landscape profiling for the development of a superior PET depolymerase.** (Top) A research flowchart is shown, indicating steps (in numerical order) for using a sequence database to assess the fitness landscape of naturally occurring PET-degrading enzymes in the polyesterase-lipase-cutinase family. (Bottom) The development of the most efficient PET depolymerase through this landscaping is summarized (from left to right) in the four panels.



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# Landscape profiling of PET depolymerases using a natural sequence cluster framework

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Enzymes capable of breaking down polymers have been identified from natural sources and developed for industrial use in plastic recycling. However, there are many potential starting points for enzyme optimization that remain unexplored. We generated a landscape of 170 lineages of 1894 polyethylene terephthalate depolymerase (PETase) candidates and performed profiling using sampling approaches with features associated with PET-degrading capabilities. We identified three promising yet unexplored PETase lineages and two potent PETases, Mipa-P and Kubu-P. An engineered variant of Kubu-P outperformed benchmarks in terms of PET depolymerization in harsh environments, such as those with high substrate load and ethylene glycol as the solvent.

Microbial decomposers can degrade newly encountered materials through enzymes with nonspecific activity, and genomic mutations that improve this emergent activity can be passed on to subsequent generations. The isolation of *Ideonella sakaiensis*, which can consume polyethylene terephthalate (PET) as a primary carbon and energy source (1), exemplifies this evolutionary concept. However, the biochemical processes used to handle such artificial substances often exhibit substrate promiscuity and provide lower fitness compared with natural substrates, making them difficult to discover. For this reason, enzyme candidates that have evolved to mediate reactions, such as the breakdown of plastics, remain a hidden part of nature's repertoire.

Biocatalysts have been widely investigated as a means to break down the ester bonds of PET and regenerate its monomers (2,3). In the 2000s, the extracellular proteins from microorganisms capable of degrading synthetic polyester—such as cutinase from the fungus *Humicola insolens* (HiC) (4) and hydrolases and cutinases from *Thermobifida* spp. (5–7)—have raised the possibility of biological PET degradation by means of  $\alpha/\beta$  structure hydrolytic enzymes. In the 2010s, other PET hydrolases, such as leaf-compost cutinase (LCC) (2,8) and PETase from *I. sakaiensis* (IsPETase) (1), were identified through homology searches in a genetic pool of estero-lytic samples. Advances in enzyme engineering (2,9–15) have been focused on IsPETase and

LCC. Sequence database searches have also revealed related local clusters of natural enzymes with PETase activity (16–21). However, these one-by-one and/or local mining approaches are limited in that they cannot see the entire fitness landscape, which provides perspective into the evolutionary success of naturally occurring PETases. A comprehensive framework based on sequence mapping of the entire genetic landscape has the potential to uncover superior PET depolymerases.

## Landscaping of polyesterase-lipase-cutinase family

To organize the landscape of PETase candidates (Fig. 1), we first constructed an initial set of 25,418 nonredundant sequences (table S1 and data S1). This set was distributed across all blocks of the ESTHER database (22) and contained various families of  $\alpha/\beta$  hydrolases, including polyesterase, lipase, tannase, fungal cutinase, type-B carboxylesterase, and fold-3 domain-containing esterase. Because 19 reported PET hydrolases with relatively high activity were in the polyesterase-lipase-cutinase family, we reduced the initial set to a subset library with 2064 sequences (Fig. 1 and data S2). The library mostly contained sequences derived from microorganisms in the phyla *Actinomycetota* and *Proteobacteria*, which accounted for approximately 83.5 and 12.5% of the sequences, respectively (fig. S1 and data S3). The lowest pairwise sequence identity in the library was 14.18% (fig. S2). Because the library was expected to cover the major potential sequences with a solid polyester depolymerization function, we referred to it as the ester-based plastic hydrolase library.

As was done with sequence similarity networks and embedding approaches (23–25), sequence homology was used to map sequences into a two-dimensional (2D) topology. How-

ever, a varying distance histogram pattern was observed for each sequence from the calculations by using Wilbur and Lipman distances (26) (fig. S3). The result suggests that setting individual relationship cutoffs might lead to a more optimized semantic network, which is similar to how individuals have their own standards for establishing social relationships. We then devised a neighborhood analysis module, which controls the network stringency from distance histogram data for each sequence (fig. S4). This approach generated a semantic network of 170 connected components (Fig. 1, and data S4 and S5), and the Prefuse force-directed-cl algorithm of Cytoscape (27) laid out the network to provide a comprehensive view of relationships. The 2064 sequences in the library were also merged into 1894 nodes, representing nonredundant natural sequences disregarding differences caused by artificial processes (data S3). Lastly, a landscape was generated for the 1894 distinct sequence indexes (N1 to N1894) across 170 clusters (C1 to C170) (Fig. 2 and data S6). As an alternative to our analysis, we used the EFI-EST tool (24) on the same set of sequences. This approach clustered some highly distinct sequences, but it was not suitable for landscaping because of the entanglement within large clusters that require high stringency (fig. S5). By contrast, our clustering reduced the number of network edges from 2064<sup>2</sup> to 14,607 and the mean and SD of the distance from 0.556 and 0.162 to 0.081 and 0.074, respectively (fig. S6), while providing high-resolution insight into intracenter relationships (Fig. 2). The number of nodes on each cluster varied widely; for example, the largest cluster (C1) contained 689 nodes, whereas 23 clusters had only one node (Fig. 2, fig. S6, and data S3). Three clusters represented the three well-studied PET hydrolase groups: (i) C20 for *Thermobifida* enzymes (assigned to Group 7 by Erickson *et al.*) containing TfCut2 (28) and *T. cellulosilytica* cutinase (Thc\_Cut2) (29), (ii) C33 for type IIB enzymes containing IsPETase (1) and RgPETase (30), and (iii) C120 for extreme thermophilic enzymes containing LCC (8) and bacterium HR29 PETase (BhrPETase) (31) (Fig. 2). Other reported enzymes were located in 22 clusters (Fig. 2 and data S3). In terms of activity on plastic substrates and catalytic properties, 145 of 170 clusters remain unexplored.

## Primary screening for PETases using stratified sampling

A primary screening for PETases was conducted by using representative nodes from 164 clusters. Because clusters with a larger number of nodes tend to have a higher mean distance (Fig. 2 and fig. S6), stratified sampling was conducted proportional to the network density on each cluster, leading to a total of 232 nodes from 164 clusters being selected for the primary screening. When the representative nodes were

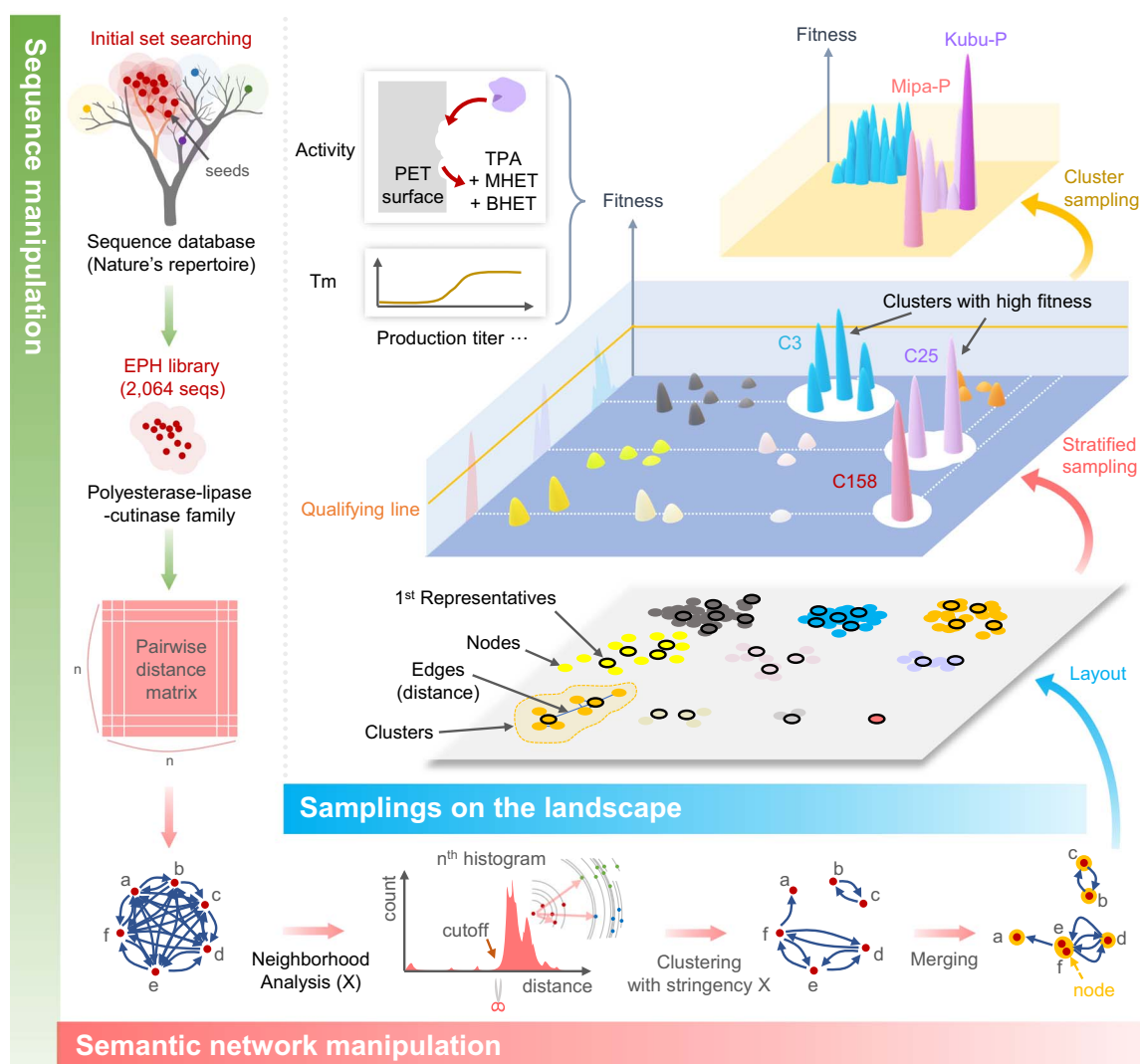
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### Fig. 1. Landscape and profiling procedures.

The ester-based plastic hydrolase (EPH) library was prepared using seed sequences within specified search ranges. The landscape method proceeds as follows: pairwise distance calculation, neighborhood analysis, and layout. The primary and secondary screenings are respectively conducted with stratified and cluster samplings (representatives) on the map to trace the fitness landscape of this protein family. TPA, terephthalic acid; MHET, mono(2-hydroxyethyl) terephthalate; BHET, bis(2-hydroxyethyl) terephthalate.



expressed in an *E. coli* strain, many nodes were not successfully produced at a level suitable for experimentation; thus, only 105 nodes were retained for participation in the primary screening. The 105 nodes contained 13 benchmarks, including the highlighted PET hydrolases such as IsPETase (C33-N1262), LCC (C120-N971), Thc\_Cut2 (C20-N6), and CaPETase (C123-N1110). Screening was conducted on the basis of evaluation criteria for three key factors: (i) monomer product release from bottle-derived PET powder at 30°C (30) for PET hydrolysis activity, (ii) protein melting temperature ( $T_m$ ) for thermal stability, and (iii) protein production titer (Fig. 3A, figs. S7 and S8, and data S3). Fifty-eight nodes (55.2%) exhibited detectable amounts of released monomer from bottle-derived PET powder, of which 34 (32%) had activity levels higher than Est2 (32), which is the PET hydrolase with the lowest activity among the active benchmarks (data S3). Of the nodes exhibiting PET hydrolysis activity, five nodes exhibited higher activity than IsPETase (Fig. 3A). The highest activity was found in N1319 from C158, which

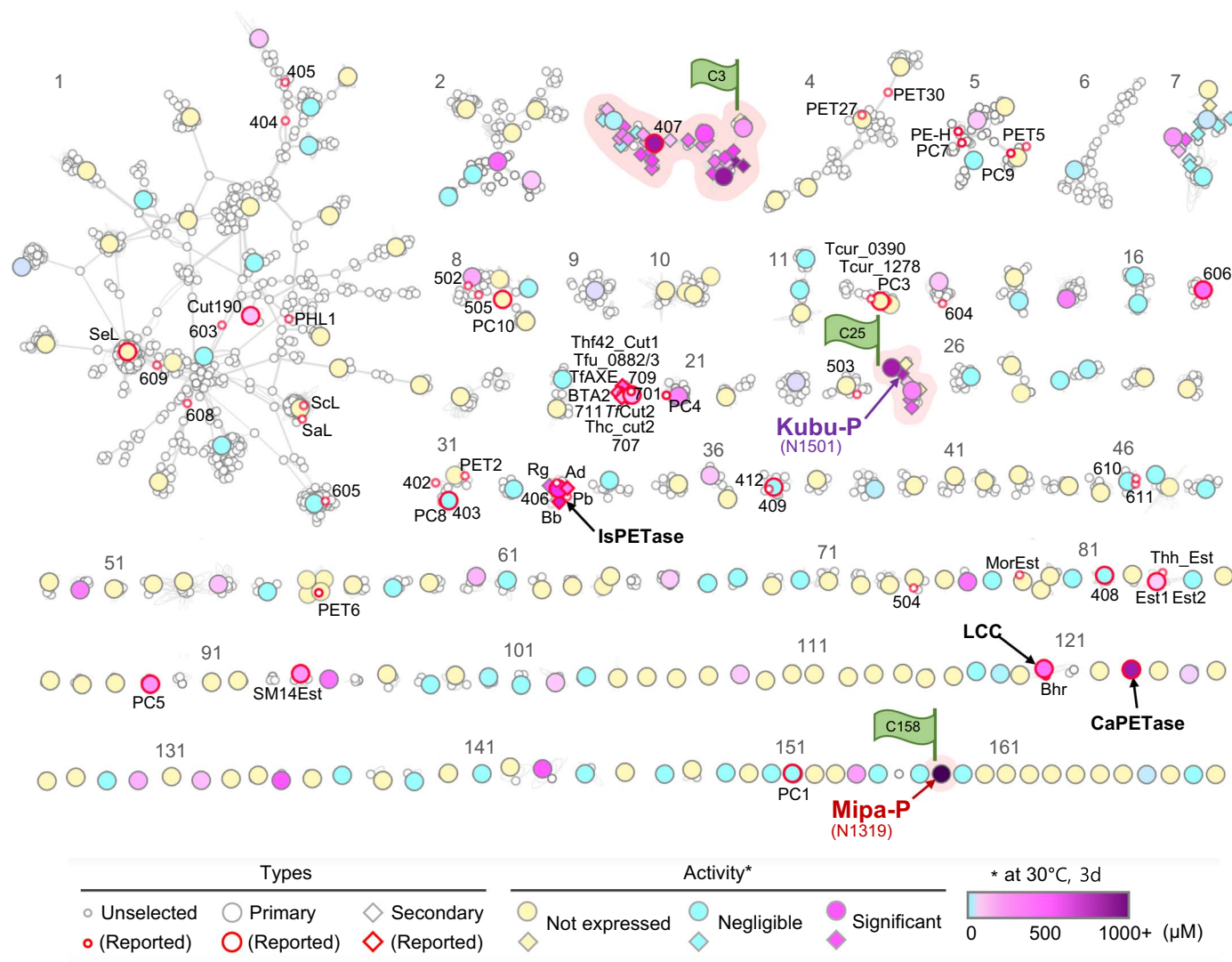
showed even higher activity than CaPETase (21) (Fig. 3A). The  $T_m$  values of 94 nodes were successfully determined (Fig. 3A, fig. S8, and data S3). There were 15 thermophilic nodes with  $T_m$  values above 60°C, and the top six nodes included four benchmarks: LCC (86.2°C), Thc\_Cut2 (69.5°C), 407 (67.7°C) (18), and CaPETase (64.9°C). The protein production titer values exhibited a large variation; SM14est (33), CaPETase, Thc\_Cut2, and Est2 were in the top quartile, whereas PC8 (21), 408 (18), LCC, 606 (18), and IsPETase were in the bottom quartile (fig. S7 and data S3).

The primary screening for PETases suggests that there are many enzymes acting on PET in natural sequences, and we identified clusters containing nodes satisfying both high PET hydrolysis activity and thermal stability. Five nodes, including N1484 and N1363 from C3, N1533 from C25, N1110 from C123, and N1319 from C158, were selected as qualified PETase candidates (Fig. 3A). The four clusters containing the qualified representative nodes (C3, C25, C123, and C158) were thus identified. Of the four clusters, C123, which contains two nodes

(N1110 and N1124; CaPETase and its analog), was the only cluster with a previously reported PETase, and the remaining three clusters (C3, C25, and C158) have been unexplored to date. N1319 from C158 exhibited the highest activity at 30°C among the tested nodes and also had a high thermal stability with a  $T_m$  of 68.6°C (Fig. 3A and fig. S8).

### Secondary screening for PETases using cluster sampling

In three unexplored clusters, C3 and C25 contained a total of 65 and 13 nodes, respectively, whereas C158 had only one node (N1319) (Fig. 2 and data S3). To identify the most talented node within C3 and C25 and assess the representativeness of the node selected in the primary screening on each cluster, we conducted a secondary screening for PETases using cluster sampling and selected 33 and 6 additional nodes from C3 and C25, respectively (Fig. 3B), which covered more than half of the untested nodes of those clusters. To further assess the representativeness of the node on each cluster, we also included 5 and 3 nodes from C20 and



**Fig. 2. Landscape profiling of PETase candidates.** Each node represents a specific natural sequence (total number of nodes is 1894). The tested nodes are colored according to their mean PET-degrading activities at 30°C, with edges shown in light-gray color. The three unexplored clusters are highlighted with green flags, and the two potent PETases Mipa-P and Kubu-P are indicated by arrows and labeled with node numbers. Nodes of benchmarks are indicated by arrows and labeled accordingly.

C33, which are well-known clusters containing Tfcut2 and IsPETase, respectively, and 10 nodes from C7, a cluster with low activity and a medium number of nodes, in the secondary screening (data S3). Overall, a total of 57 nodes from five clusters participated in the secondary screening. The success rate of protein production increased dramatically compared with the primary screening, with 53 nodes (92.9%) producing at a level suitable for the experiment (fig. S7 and data S3), indicating that the protein expression patterns of the nodes were quite similar within a cluster. The PET hydrolysis activity and Tm values were also substantially higher, with 49 nodes (92.4%) exhibiting detectable amounts of released monomer and 30 nodes (56.6%) having Tm values >60°C (fig. S8 and data S3). Nodes from C3 and C25 had a sample mean for PET hydrolysis

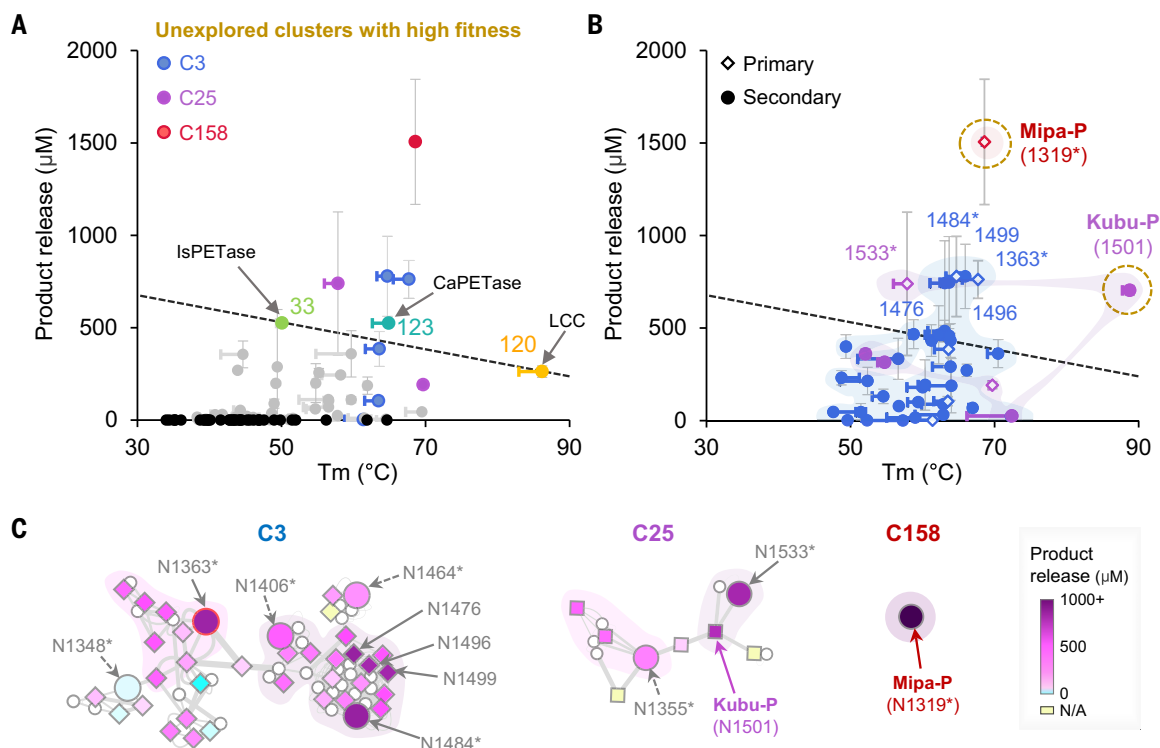
activity or Tm values that were higher than those from C20 and C7, or C33, respectively (fig. S9). These results suggest that the representative nodes selected for the primary screening were reasonable and that PET degradation fitness is connected within each cluster. However, we also observed disparities in regional characteristics within individual clusters. In C3, local regions with N1484 or N1499 had high activity, whereas those with N1348 tended to be relatively weak (Fig. 3C). Similarly, C25 was also divided into small local regions with varying activity or Tm values (Fig. 3C and fig. S10). These observations indicate that, although the nodes on each cluster share relatively similar enzymatic characteristics, the PETase candidates could not be completely separated into clusters with distinct characteristics, a factor that emphasizes the importance

of density-based stratified sampling in enzyme screening.

As a result of the secondary screening, four additional qualified nodes (N1501 from C25 and N1476, N1496, and N1499 from C3) were identified, approaching a final total of eight qualified nodes, including the four nodes found in the primary screening (N1319 from C158, N1533 from C25, and N1363 and N1484 from C3) (Fig. 3, B and C). The mapping results of the two screening procedures distinguished three clusters with high PETase fitness and showed the representativeness of the nodes by their positions, demonstrating the success of the landscaping method used in this study (Fig. 2). Thus, our landscaping-and-sampling approach presents a successful model of fitness landscape for accumulating sequence-to-activity data.

**Fig. 3. PETase screening.**

**(A)** The primary screening for PETase using stratified sampling. Nodes releasing detectable monomers from PET and those with negligible activity are shown as gray and black circles, respectively. Standard deviation (SD) of the activity data are presented as bars ( $n = 2$  to  $\sim 11$  as shown in data S3). The melting temperature ( $T_m$ ) values are shown as circles, linked with a line to corresponding Boltzmann  $T_m$  values represented as bars. IsPETase, LCC, and the nodes in the three unexplored clusters are colored according to the color schemes shown in the panel. **(B)** Secondary screening for PETase using cluster sampling. Nodes in C3, C25, and C158 are shown on an activity- $T_m$  plot. The eight qualified nodes are labeled, and the representative nodes in the primary screening are distinguished by the marker shape and by adding an asterisk to the label. The shadows represent edge connections between the nodes. The connection line between IsPETase and LCC is shown as a dotted line. **(C)** Network of the three unexplored clusters. The figure is drawn as in Fig. 2. The qualified nodes are indicated by solid-line arrows, and the nodes in the primary screening but not selected as qualified are indicated by dashed-line arrows.



### Two high-fitness nodes and comparison with benchmarks

To select the most potent enzymes among the eight nodes discovered from the primary and secondary screenings, we conducted a final evaluation by measuring the PET hydrolysis activity at various temperatures using amorphized postconsumer PET powder (21). Of the eight nodes, two (N1319 and N1501) exhibited outstanding PET hydrolysis activity across the range of temperatures tested: N1319 demonstrated the highest product release at 40°C, and N1501 exhibited the highest product release at temperatures ranging from 50° to 70°C (figs. S11 and S12). Taken together, of 158 nodes that participated in the three screening rounds, N1319 from C158 and N1501 from C25 were selected as the most potent nodes. Because the genes coding for N1319 (accession code: WP\_091558835.1) and N1501 (accession code: WP\_116180173.1) originated from microbial strains of *Microspora pattaloongensis* DSM 45245 (34) and *Kutzneria buriramensis* DSM 45791 (35), we refer to these PET depolymerases as Mipa-P and Kubu-P, respectively.

We compared the enzymatic characteristics of Mipa-P and Kubu-P with those of the benchmarks CaPETase, IsPETase, and LCC. For a detailed comparison, we investigated initial rate and depolymerization extent as indicators of PET

hydrolysis speed and duration of the enzymes. The apparent initial rate and the depolymerization extent were calculated by measuring the PET hydrolysis rate during the initial 3 hours and the total amount of monomers released after 168 hours, respectively, at temperatures ranging from 30° to 70°C under the screening condition of 1.48% (w/w) PET load and 500 nanomolar enzyme concentration ( $\sim 1 \text{ mg}_{\text{enzyme}} \text{ g}_{\text{PET}}^{-1}$ ) (Fig. 4A). In these experiments, the initial rate tended to increase with higher temperatures but started to decrease beyond a certain temperature close to  $T_m$  for Mipa-P ( $T_m = 68.6^\circ\text{C}$ ), CaPETase ( $T_m = 64.9^\circ\text{C}$ ), and IsPETase ( $T_m = 50.1^\circ\text{C}$ ) (Fig. 4B). This pattern of increase for initial rate according to the temperature was similar for all enzymes, indicating that enzymes that are generally more active at low temperature will still exhibit more rapid activity at higher temperatures within their tolerance range.

Of the five PETases tested, Mipa-P exhibited the highest initial rate except for 60° and 70°C, where it was outperformed by Kubu-P, which has a much higher  $T_m$  than Mipa-P (Fig. 4, A and B). Kubu-P also had a much higher initial rate than the benchmarks at all temperatures. In terms of depolymerization extent, Kubu-P exhibited the best performance at most temperatures, followed by Mipa-P and LCC, which

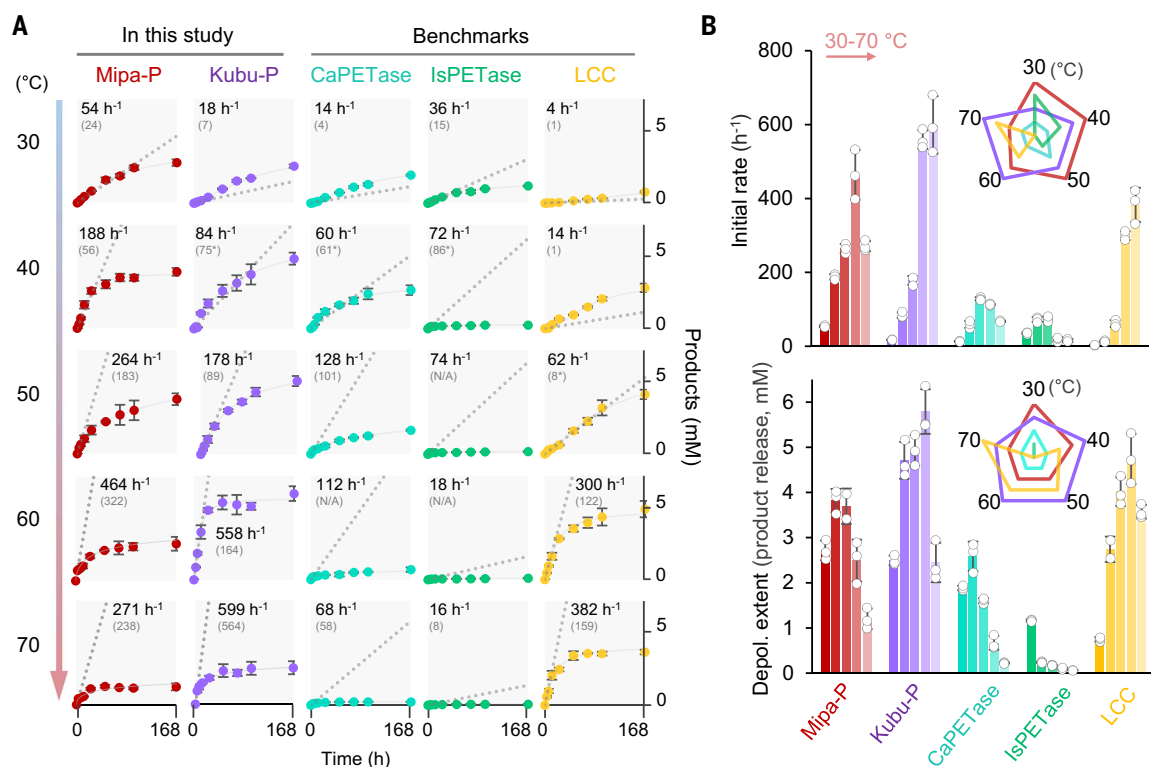
had the highest depolymerization extent at 30° and 70°C, respectively (Fig. 4, A and B). LCC generally exhibited slow-and-steady characteristics, with the lowest depolymerization extent at 30°C among the PETases tested, but this increased with higher temperatures, reaching the highest depolymerization extent at 70°C (Fig. 4B). These results indicate that the two PETases Mipa-P and Kubu-P, discovered in the PETase screenings described in the present study, are potent PETases that demonstrate a high decomposition speed and duration when compared with benchmarks.

### Mipa-P<sup>M19</sup> and Kubu-P<sup>M12</sup> variants

We determined crystal structures of Mipa-P and Kubu-P (table S2) and performed mutagenesis experiments to understand key structural features of these enzymes. Both proteins share most of the local subnetwork structures of CaPETase (21), but they do not have the wobbling tryptophan mechanism of IsPETase (36) (figs. S13 to S17 and supplementary text). To improve the thermal stability of these enzymes while maintaining or increasing their activity levels, we aimed to introduce diversity through various sequences from the nodes characterized in this study. This approach, which we termed cross-template engineering, was expected to minimize extreme variations that

**Fig. 4. Comparison of Mipa-P and Kubu-P with benchmarks.**

The comparison results with the benchmarks are shown. **(A)** The initial rate and depolymerization extent of Mipa-P and Kubu-P are compared with those of benchmarks. Time course activities of the enzymes are measured at temperatures ranging from 30° to 70°C. The initial rates correspond to the apparent velocity for 3 hours and are indicated by a gray dashed line with their values. The values for 1-hour initial rates, derived from nonlinear fits using the four-point logistic model, are also labeled in parentheses (asterisks indicate nonweighted fit because the weighted model was not converged). Data are presented as mean values  $\pm$  SD ( $n = 3$ ). **(B)** Initial rate (3 hours) and depolymerization (Depol.) extent (168 hours) of enzymes along temperatures. Radar charts of initial rate and depolymerization extent ranks along temperatures by enzymes are also displayed. The ranks from 1 to 5 are shown with axes in direction toward the center.

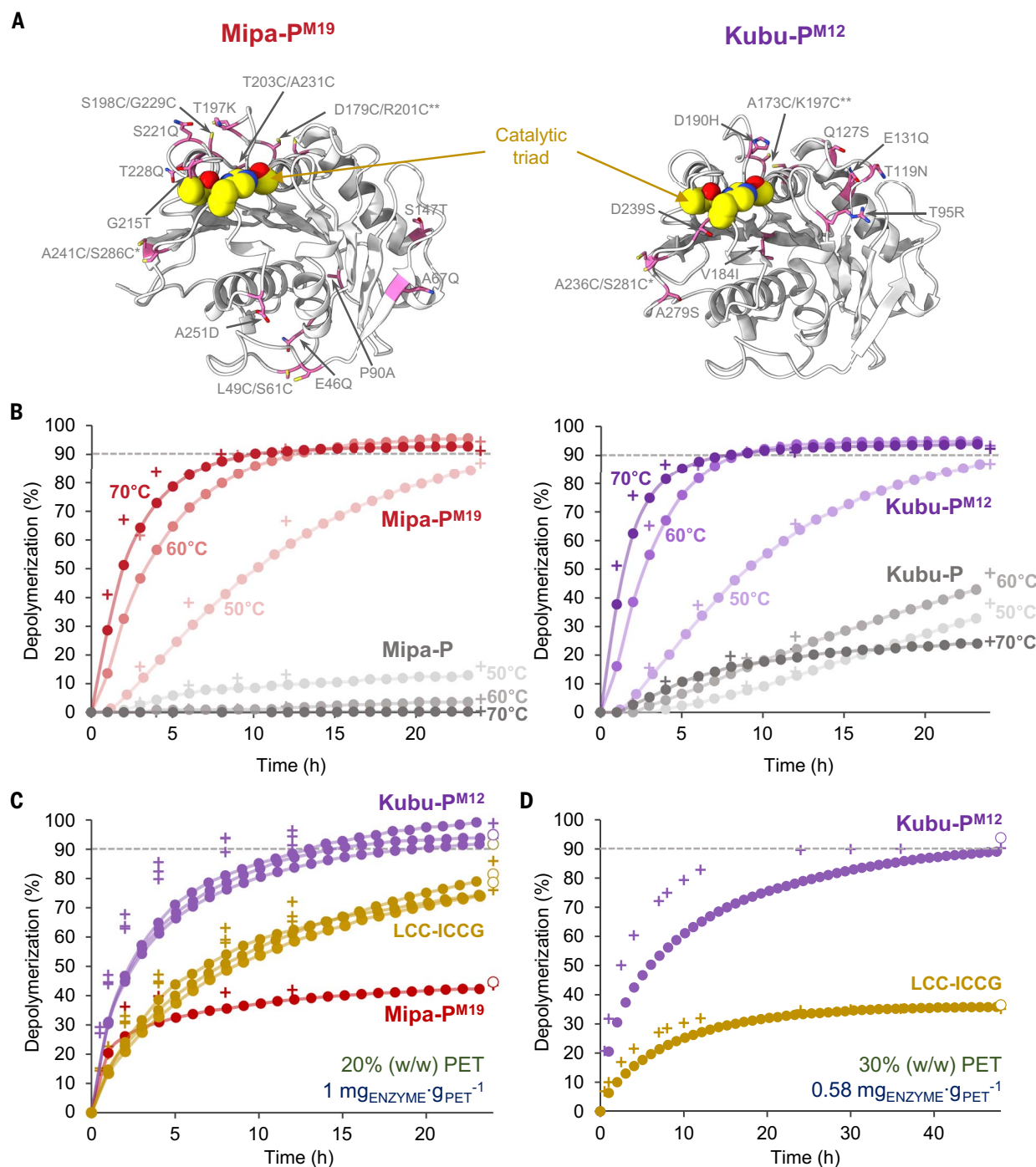


could be detrimental to enzyme activity because it exclusively utilizes conserved residues found in natural sequences. After aligning representative sequences from the qualified clusters with Mipa-P and Kubu-P sequences, we rationally selected specific structural positions for modification, where amino acids present at corresponding sites across the alignment served as mutation candidates (fig. S18). On the basis of the cross-template information, we selected 14 and 12 substitution candidates for Mipa-P and Kubu-P, respectively. Additionally, five and two candidate positions were selected for potential disulfide bond introduction, resulting in final totals of 19 and 14 replacement candidates for each enzyme, respectively (figs. S19 and S20). Mutation points exhibiting non-negative effects for  $T_m$  were serially accumulated and the activity of the intermediate variants was evaluated (figs. S19 and S20) to produce the Mipa-P<sup>M19</sup> and Kubu-P<sup>M12</sup> variants, with a total of 19 and 12 amino acid substitutions (Fig. 5A). The thermal stability of these two variants was higher, with Mipa-P<sup>M19</sup> achieving a  $T_m$  of 92.4°C, representing an increase of 24.1°C compared with the wild-type counterpart (figs. S19 and S21). For Kubu-P<sup>M12</sup>, its  $T_m$  was not measurable within the temperature range of 0 to 99.9°C at a scanning rate of 0.1°C·s<sup>-1</sup>, indicating that its  $T_m$  may be >99.9°C (figs. S20 and S21). Both the initial rate and depolymerization extent of

Mipa-P<sup>M19</sup> and Kubu-P<sup>M12</sup> were also higher at both 50° and 70°C compared with their wild-type versions (fig. S22), illustrating the enhanced enzymatic properties of these variants. The effective mutations introduced disulfide bonds, salt bridges, and hydrogen bonds (figs. S23 and S24). In particular, the D190H (D190→H) mutation in Kubu-P<sup>M12</sup> on the active site loop, corresponding to the D186H mutation in IsPETase-EHA (9), was highly effective in stability, reaffirming the importance of this loop (fig. S24 and supplementary text).

To investigate the performance of Mipa-P<sup>M19</sup> and Kubu-P<sup>M12</sup> in a pH-stat reaction that allows nearly complete depolymerization (2), we compared them with their wild-type counterparts in a bioreactor with reaction conditions of pH 8.0 (figs. S12 and S25), 2% (w/w) PET powder, and 3 mg<sub>enzyme</sub>·g<sub>PET</sub><sup>-1</sup> enzyme at 50° to 70°C. The improved performance of these two variants in the agitated pH-stat bioreactor was more striking than in the closed reaction (Fig. 5B). Whereas both wild-type enzymes achieved less than 50% depolymerization after 24 hours, the engineered variants degraded more than 90% of the PET within 24 hours at all temperatures tested (Fig. 5B). Mipa-P<sup>M19</sup> was slightly faster than Kubu-P<sup>M12</sup> at 50°C after 12 hours, but at 60° and 70°C, the depolymerization rate of Kubu-P<sup>M12</sup> was higher, with depolymerization of 51.2 and 75.8% after 1 and 2 hours at 70°C,

respectively (Fig. 5B). Under the same bioreactor conditions, Hot-PETase (15) exhibited optimal performance at 60°C, achieving a final degree of depolymerization of 87% after 24 hours. However, its performance decreased at 70°C, resulting in a final degree of depolymerization of only 17% and showing the lowest durability among those tested. LCC-ICCG (2) achieved a final degree of depolymerization of 90% at 60 and 70°C but could not reach 90% at 50°C, possibly because its depolymerization activity was lower than that of Mipa-P<sup>M19</sup> and Kubu-P<sup>M12</sup> (fig. S26). These results indicate that the two engineered variants, Mipa-P<sup>M19</sup> and Kubu-P<sup>M12</sup>, have excellent PETase characteristics, with Kubu-P<sup>M12</sup> in particular exhibiting both the highest depolymerization activity and highest final degree of depolymerization of the PETase variants tested. However, the depolymerization reaction using nonamorphized PET powder from the postconsumer flake showed that all these variants are more optimized to amorphized PET powder (fig. S27). When the enzyme kinetics of Kubu-P<sup>M12</sup> was analyzed by varying the enzyme concentration and PET substrate loads, the enzyme concentration in the experiments of 2% (w/w) PET condition approximated the concentration for maximum productivity, but the substrate concentration for the optimal initial rate was found to be >20% (w/w) (fig. S28), which represents highly productive industrial conditions.



**Fig. 5. Performances of Mipa-P<sup>M19</sup> and Kubu-P<sup>M12</sup>.** (A) Crystal structures of Mipa-P<sup>M19</sup> and Kubu-P<sup>M12</sup>. The structures are shown as gray cartoon models, and the catalytic triad is highlighted by yellow sphere models. Mutations relative to the starting protein sequences are distinguished by magenta stick models and labeled. Single-letter abbreviations for the amino acid residues are as follows: A, Ala; C, Cys; D, Asp; E, Glu; G, Gly; H, His; I, Ile; K, Lys; L, Leu; N, Asn; P, Pro; Q, Gln; R, Arg; S, Ser; T, Thr; and V, Val. (B) Comparisons of Mipa-P<sup>M19</sup> and Kubu-P<sup>M12</sup> with their wild-type

counterparts. The PET depolymerization experiments were performed in a pH-stat bioreactor. The graphs of linked circles or crosses indicate degree of depolymerization calculated by monomer release (HPLC sampling) or ester bond cleavage (NaOH consumption), respectively. The level of 90% degree of depolymerization is indicated by a gray dashed line. (C and D) Depolymerization performance by the enzymes in high-substrate load conditions. The bioreactor operation with PET loads of 20% (C) and 30% (D) are drawn as in Fig. 5B. The final weight loss for each reaction is also shown as a white circle.

#### High tolerance to high PET loads of Kubu-P<sup>M12</sup>

Industrial conditions often represent a harsh environment for enzymes owing to the high temperatures, high substrate loads, and use of organic solvents to maximize productiv-

ity (fig. S29). Enzymes with a higher activity than LCC-ICCG under mild conditions exhibit a low final degree of depolymerization when using a high substrate loading of 16.5% (w/w) (37). Thus, we evaluated the tolerance

of Mipa-P<sup>M19</sup> and Kubu-P<sup>M12</sup> to high substrate loads with reaction conditions of 20% (w/w) PET and 1 mg<sub>enzyme</sub>·g<sub>PET</sub><sup>-1</sup> at 70°C in comparison with LCC-ICCG. Of the three engineered variants, Kubu-P<sup>M12</sup> showed the highest degree

of depolymerization, reaching 45.2% in the first hour, approximately 1.5 and 3.5 times as high as that of Mipa-P<sup>M19</sup> and LCC-ICCG, respectively (Fig. 5C). Kubu-P<sup>M12</sup> also achieved a degree of depolymerization of 90% within 8 hours. However, the activity of Mipa-P<sup>M19</sup> slowed after 2 hours, and the final decomposition rate at 24 hours remained at approximately 40% (Fig. 5C). LCC-ICCG maintained its activity more consistently than Mipa-P<sup>M19</sup>, but its final depolymerization rate at 24 hours was still limited to 80% (Fig. 5C). The difference in the PET degradation behavior of Kubu-P<sup>M12</sup> and LCC-ICCG was confirmed with two biologically independent bioreactor trials. Under harsher conditions, with a PET loading of 30% (w/w) and an enzyme supplement of 0.58 mg<sub>enzyme</sub>·g<sub>PET</sub><sup>-1</sup>, the difference in performance between Kubu-P<sup>M12</sup> and LCC-ICCG was even greater. Kubu-P<sup>M12</sup> depolymerized 90% of the PET in 24 hours, whereas LCC-ICCG achieved <40% depolymerization after 48 hours (Fig. 5D), indicating that Kubu-P<sup>M12</sup> performs better at higher substrate loading. On the basis of these results, we conclude that Kubu-P<sup>M12</sup> is an excellent PETase variant in terms of both depolymerization speed and duration when compared with other engineered PETase variants.

### Enzyme-catalyzed PET glycolysis using Kubu-P<sup>M12</sup>

We then tested whether Kubu-P<sup>M12</sup> could maintain its structure in ethylene glycol (EG) for enzyme-catalyzed PET glycolysis (38), which can produce bis(2-hydroxyethyl) terephthalate (BHET) through the reaction of the hydroxyl-

groups in EG with the tetrahedral intermediate formed during the deacylation reaction (Fig. 6A). PET degradation experiments were conducted in water-depleted EG (>95%) solvent at 40°C using the PETase variants Mipa-P<sup>M19</sup>, Kubu-P<sup>M12</sup>, LCC-ICCG, and Hot-PETase. High levels of BHET as the main product (>98%) were observed with the use of Mipa-P<sup>M19</sup>, Kubu-P<sup>M12</sup>, and LCC-ICCG after incubation for 24 hours, whereas Hot-PETase produced a negligible amount of BHET (Fig. 6B and figs. S30 and S31). Kubu-P<sup>M12</sup> exhibited the highest glycolysis activity, with a greater than fourfold-higher activity compared with Mipa-P<sup>M19</sup> and LCC-ICCG, which exhibited activity similar to each other (Fig. 6B). When the tolerance of these variants to EG was investigated at various temperatures, Mipa-P<sup>M19</sup> and Kubu-P<sup>M12</sup> generally maintained their glycolysis activity at 40° and 50°C for 10 days in EG, whereas LCC-ICCG was more susceptible to inactivation in EG (Fig. 6B). As observed in other protic solvent-based esterification reactions, addition of two buffer agents increased the enzymatic glycolysis activity (fig. S32). When enzyme-catalyzed PET glycolysis by Mipa-P<sup>M19</sup> and Kubu-P<sup>M12</sup> was tested in the Tris-containing EG solution in the bioreactor, the use of Kubu-P<sup>M12</sup> resulted in the highest production of BHET reported to date, producing approximately 21 mM BHET with a monomer fraction purity of 98%, which corresponded to a degree of depolymerization of 8% within 48 hours at 40°C (Fig. 6C). At 200 hours, the reaction even reached near the BHET solubility in EG at this temperature (39)

(fig. S33). These results demonstrate that Kubu-P<sup>M12</sup> satisfies the characteristics required for enzyme-catalyzed PET glycolysis, such as tolerance to EG, high temperatures and high PET loads, PET decomposition activity, and enzyme reactivity to EG as a second substrate, thus demonstrating the potential for use in enzyme-catalyzed PET glycolysis for the biocatalytic recycling of PET.

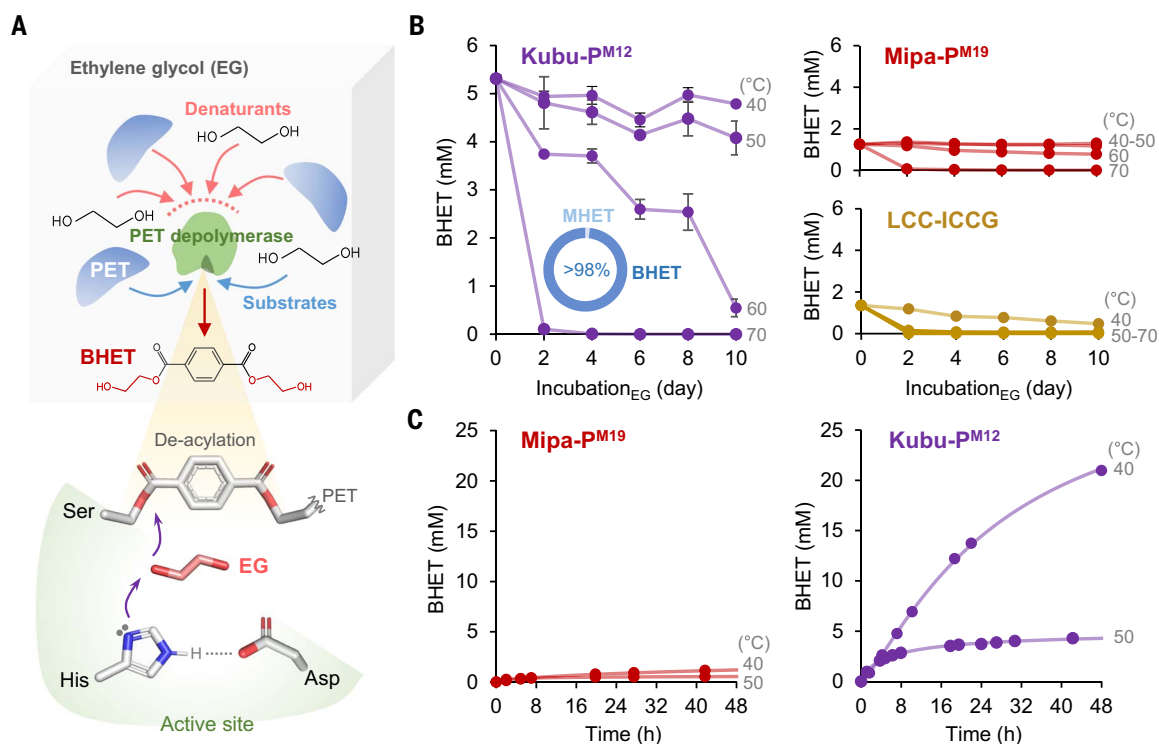
### Discussion

In this study, we investigated the polyesterase-cutinase-lipase family of ester-based plastic hydrolases through a landscape-profiling framework. Our biochemical data reveal a fluctuating surface with clustered peaks of high-activity enzymes. Although known esterases outside of this family were not considered owing to their relatively weak activity (18, 40, 41), the PETase screening in our study based on accumulated sequence data was nonetheless extensive and from geographically diverse sequence sources. The available isolation source data from BioSample provides environmental origins of many potential PETases, such as rhizosphere and mangrove soil (e.g., C2, C3, C7, C21, C25, C50, C54, C60, C66, C135, and C154), marine and water sediments (C5, C13, C16, and C76), and hot compost (C20) (data S7). Landscape profiling of this family revealed that enzymes acting on a solid-state PET substrate exist within a variety of clusters, with at least 90 nodes discovered with PETase activity and 17 benchmarks across 43 clusters. Of these templates, 82 nodes from 33 clusters

**Fig. 6. Use of Kubu-P<sup>M12</sup> in enzyme-catalyzed PET glycolysis.** (A) Schematic diagram of enzymatic glycolysis and reaction mechanism at the deacylation step.

Under the ethylene glycol (EG) solvent, both PET and EG can act as denaturants and substrates of the enzyme. (B) Enzyme-catalyzed PET glycolysis activity and inactivation patterns of the three enzymes by preincubation in EG. The mol fraction of BHET, mono(2-hydroxyethyl) terephthalate (MHET), and negligible terephthalic acid (TPA) at 1-day reaction are shown in a donut chart. The temperature condition is labeled at the right side. Data are presented as mean values ± SD (n = 3).

(C) Enzyme-catalyzed PET glycolysis performed using Mipa-P<sup>M19</sup> and Kubu-P<sup>M12</sup> in a bioreactor scale. The temperature condition is labeled as in (B).



have not, to the best of our knowledge, been previously identified as a homologous group in past studies, indicating that 90% of the templates were unrecognized (data S3). Our stratified sampling method sought to trace the fitness landscape and may provide a bird's-eye view of the natural distribution of these enzymes. We highlighted three clusters, C3, C25, and C158, that contained nodes with high PET decomposition potential; and two PETases, Mipa-P and Kubu-P, with the characteristics of potent PETases were discovered on C25 and C158, respectively.

Bacterial strains *M. pataloongensis* (34) and *K. buriramensis* (35), containing genes coding for Mipa-P and Kubu-P, respectively, were isolated from tropical deciduous areas in Thailand, an area with a rainforest climate and rich in ester bioresources. However, the genomes of these two microbes do not appear to have a 1,2-dihydroxy-3,5-cyclohexadiene-1,4-dicarboxylate dehydrogenase-coding gene, suggesting that they are not likely to assimilate terephthalate via a known aerobic metabolic pathway (1). We thus infer that the enzymes identified in our screen may have other natural substrates but can still be exploited for their PETase activity. It is also worth noting that the optimal growth temperatures for most microorganisms containing enzymes with a high  $T_m$  are not high. Although gene mining that is limited to bacteria growing at high temperatures is effective in identifying enzymes with a high thermal stability, this strategy may overlook enzymes from low-temperature host that are nonetheless effective in harsh extracellular environments.

The increase in initial rate with higher temperatures below the  $T_m$  (Fig. 4B) indicates that low-temperature activity better represents catalytic efficiency than does high-temperature activity, which is affected by the duration of catalyst activity. However, current research directions place greater weight on slow-and-steady thermophilic templates such as LCC and BhrPETase because differences in durability are more prominent above the glass transition temperature of PET (42) than are differences in catalytic efficiency between enzymes. Indeed, as shown by the IsPETase (15) and CaPETase variants (21), the strategy of increasing the durability of enzymes with high activity at low temperatures has only been partially successful, increasing the speed of PETase but not durability under harsh conditions. We believe that our discovery approach, which simultaneously evaluates low-temperature activity and thermal stability, is crucial to finding highly potent PETases such as Mipa-P and Kubu-P, which offer both high catalytic capability and thermal stability. In particular, Kubu-P appears to be a more optimal template for the engineering of a variant with a high tolerance to high temperatures, high substrate loads, and the use of water-depleted

EG as a solvent. Consequently, Kubu-P<sup>M12</sup> simultaneously achieved all the key properties required for PET degradation.

However, the degradation results obtained using intact postconsumer PET flakes indicate that these enzymes still require an amorphization process. To induce crystalline chains into reaction intermediates at temperatures utilized in enzymatic depolymerization, sufficient binding energy must be provided. Enzymes from the polyesterase-lipase-cutinase family, characterized by substrate promiscuity, may not represent the optimal scaffold for this function. We believe that degrading crystalline PET itself will require a multienzyme process involving a consortium of enzymes with random-scission and processive mechanisms through endo- and exo-depolymerases, similar to numerous natural degradation mechanisms. For this reason, a simple amorphization process would be more advantageous for industrial efficiency.

In the PET supply chain and downstream recycling processes, enzymatic glycolysis offers an economic advantage because it does not require a prior esterification process or titrants for pH adjustment, and the resulting product has a negligible water content, which reduces high latent heat exchange during purification. A previous study using the commercial enzymes *Candida antarctica* lipase B and HiC could not achieve sufficient depolymerization rates or amounts even under optimal conditions (38). By contrast, the apparent rate of glycolysis using Kubu-P<sup>M12</sup> was approximately 109 hours<sup>-1</sup> (within 48 hours) at 40°C (Fig. 6C), a speed comparable to that of hydrolysis. Nevertheless, there is still an important industrial limitation on yield due to the solubility of BHET. Further engineering of Kubu-P<sup>M12</sup> for higher EG tolerance would allow glycolysis reactions to occur at higher temperatures, enhancing both its depolymerization rate and degree. This improvement may also act as a starting point for the development of a BHET recovery process (43), by producing BHET at concentrations sufficient to form crystalline nuclei at low temperatures (fig. S34). Overall, we believe that our landscape profiling of PET depolymerases provides for greater understanding of the natural distribution of PETases and will enable more efficient enzymatic PET depolymerization through hydrolysis, glycolysis, and potentially other methods.

## Materials and methods

### Library construction, clustering, and visualization

The initial set of 25,418 non-redundant sequences was retrieved from GenBank Release 235 in December 2019 using PSI-BLAST searches with seed sequences. The value of the search threshold for each seed are summarized in table S1. The subset library of 2064 sequences was constructed by reducing the seed sequences based on the criterion of polyesterase-lipase-

cutinase family and partial gene removal. This was followed by clustering using a method optimized for landscape analysis. The clustering algorithm utilizes a pairwise distance matrix (values ranging from 0 to 1) developed by Wilbur and Lipman (26), implemented in clustalo (44). The distance data are truncated at a significant density distance to reduce meaningless relationship entanglement, thereby generating a network of edges between sequence. Briefly, the significant density distance suggests a high probability of clusterable group existence and is determined from a histogram of distances for each sequence. We refer to this heuristic procedure as neighborhood analysis. The histograms and cut-off data for all sequences are shown in fig. S3. The neighborhood analysis distinguishes itself from conventional k-mer or k-nearest neighbor (kNN) clustering, which is typically implemented in sequence similarity network analyses. It does so by utilizing flexible numbers of neighbors that leverage the complexity of distance-dependent histogram patterns for each sequence. This approach was implemented using a Python3 code with eight hyperparameters (fig. S4).

The initial nodes containing sequences identical in the conserved domain and/or artificial sequences were merged into single nodes and then indexed as N1 to N1894. This was achieved by using accession codes to retrieve entries from the Identical Protein Group (IPG) using Entrez software (45), combined with utilization of the idmapping\_selected.tab file from the UniProtKB database (24 February 2022). Therefore, each node represents a specific natural sequence origin. The resulting relationship clusters generated by the neighborhood analysis are visualized as a landscape using Cytoscape (27) software with the Prefuse force-directed-cl layout. This layout provides forces to position sequences relative to the network structure. The layout parameters and edge weight settings are summarized in table S3. After layout, self-connecting edges were removed to finalize the landscaping. For the enzyme similarity tool EFI-EST (24), we generated networks by varying the alignment score thresholds using the fasta format of the 2064 sequences via the web server (<https://efi.igb.illinois.edu/efi-est/>).

The ID value of the library, cluster numbers, node numbers, accession codes, and other analytical and experimental data are summarized in data S3. The landscape of the network data as a Cytoscape (.cys) format is also available in data S6. The phyla of each bioresources were analyzed using taxonomy data from NCBI, including genus and species names, accessed via the \_search\_scientific table of The Catalogue of Life (COL) database (46). For Biosample (47) analysis, we accessed the database using accession codes of each nucleotide sequences in the IPG database through Entrez.

### Preparation of genetic sources

Genetic information can be found in data S3. For cloning, we obtained double-stranded DNAs from synthetic sources (codon-optimized for *E. coli*, Twist Bioscience) or genomic sources (*Marinactinospora thermotolerans*, *Roseateles depolymerans*, and *C. aurantiacum*). The genes were amplified either by DH10B strain culture or polymerase chain reaction (PCR) using oligonucleotides listed in table S4. Subsequently, they were inserted into commercial vectors, pET21b, pET22b, or pET30a, using restriction enzymes NdeI and XhoI. Signal peptides of the genes were predicted using SignalP 5.0 (48). The expressed protein constructs, along with secondary structure prediction and signal peptide information, were designed and recorded in data S3. The pET21b-based plasmids for LCC-ICCG and Hot-PETase were synthesized (table S5) and subcloned using the same enzymes. Site-directed mutagenesis for variants was performed by PCR using oligonucleotides listed in table S4, and all variants were verified by Sanger sequencing.

### Recombinant protein expression and purification

The constructed plasmids were transformed into the *E. coli* strain BL21(DE3)-T1<sup>R</sup>. To help formation of disulfide bonds, variants of Mipa-P and Kubu-P (Mipa-P<sup>M19</sup> and Kubu-P<sup>M12</sup>), as well as LCC-ICCG, and Hot-PETase, were produced by cultivating Rosetta gami-B strain transformed with their respective plasmids. The transformed cells were cultured in 1 liter of lysogeny broth (LB) medium containing either 100 mg liter<sup>-1</sup> Ampicillin or 50 mg liter<sup>-1</sup> Kanamycin, depending on the antibiotic resistance conferred by each vector, at 37°C. When the optical density of the culture broth reached OD<sub>600</sub> of 0.6, 0.5 mM isopropyl β-D-1-thiogalactopyranoside (IPTG) was added to induce protein expression in the T7-LacO promoter system. After further culturing for 20 hours at 18°C with shaking at 120 rpm, the cells were harvested by centrifugation at a relative centrifugal force (RCF) of 4,000 ×g for 15 min at 20°C.

The cell pellet was resuspended in lysis buffer (40 mM Tris-HCl, pH 8.0), and cell lysis was performed using ultrasonication. The cell lysate was centrifuged at an RCF of 13,000 ×g for 30 min, and the supernatant was collected. For affinity chromatography, the supernatant was applied to a Ni-NTA agarose column (Qiagen) that had been equilibrated with the lysis buffer. After washing the column with the lysis buffer containing 30 mM imidazole, the bound proteins were eluted using 300 mM imidazole in the lysis buffer. All protein purification steps were conducted at 4°C. The purities of the eluted proteins were analyzed by sodium dodecyl sulfate-polyacrylamide gel electrophoresis (SDS-PAGE) and Coomassie blue staining, as shown in fig. S6. The protein production titer of each protein was quantified using a BioTek Epoch microplate

spectrophotometer and Gen5 microplate data analysis software. For the primary, secondary, and final screenings, the produced enzyme solutions were concentrated using an Amicon Ultra Centrifugal filter unit (10,000 Da, Millipore, Billerica, MA, USA) to a final concentration of 1 mg ml<sup>-1</sup>. For the hydrolysis reactions in the bioreactor, the solutions were concentrated to 5 mg ml<sup>-1</sup> and then used in the reaction. To avoid precipitation of Mipa-P<sup>M19</sup> and LCC-ICCG during concentration, sodium chloride was added to the eluted solution to achieve a concentration of 150 mM. For the feasibility tests and bioreactor experiments for enzyme-catalyzed PET glycolysis, we prepared enzyme solutions at concentrations of 10.2 and 20 mg ml<sup>-1</sup>, respectively. For crystallization, further purification through size exclusion chromatography was conducted using a HiPrep 26/60 Sephacryl S-100 HR column (320 mL, GE Healthcare Life Sciences) equilibrated with the lysis buffer. The eluted proteins were similarly concentrated to 10 mg ml<sup>-1</sup> and prepared for next crystallization steps.

### PET powder preparation

The number-average (Mn), weight-average (Mw), highest peak (Mp) molecular weights, and the average diameter of the PET powders used were analyzed using gel permeation chromatography and a laser diffraction particle size analyzer, respectively. These analyses were certified by the Korea Polymer Testing & Research Institute (KOPTRI). The microcrystalline PET powder, pulverized from a commercial bottle and prepared as described in Sagong *et al.* (30) (average diameter of 182.4 μm, 8,800 Mn, 23,800 Mw, 19,900 Mp, and 17.4% crystallinity), was used for the primary and secondary PETase screenings. The amorphized postconsumer powder (average diameter of 247.8 μm, 7,060 Mn, 16,700 Mw, 13,400 Mp, and 9.9% crystallinity) from transparent flakes (Yoochang R&C, Republic of Korea), prepared as described in Hong *et al.* (21), with 200 μm sieving, was subsequently used for final screening, further comparisons, bioreactor, and enzyme-catalyzed PET glycolysis experiments. The crystallinity values were estimated from differential scanning calorimetry (DSC) data from the previous articles. The non-amorphized PET powder was prepared by pulverizing the reclaimers' transparent flake (Yoochang R&C, Republic of Korea) with 300 μm sieving. Its crystallinity was calculated as 23.6% (fig. S27).

### PETase activity assays

To measure PETase activity for the screenings, 15 mg of PET powder was soaked in 1 ml reaction solution containing 50 mM glycine-NaOH (pH 9.0) and 500 nanomolar enzyme (approximately 13–16 μl of 1 mg ml<sup>-1</sup> enzyme solution). The primary and secondary screenings were performed at 30°C for 3 days, but final screening was conducted at temperatures ranging from

30° to 70°C for 12 hours. For the comparison of time-course activity (table S6), the reactions were conducted for 3, 6, 12, 24, 48, 72, 96, or 168 hours at temperatures ranging from 30° to 70°C. All the reaction was terminated by removing the substrate using a PVDF syringe filter (0.22 μm), and the main products, terephthalic acid, mono(2-hydroxyethyl) terephthalic acid, and bis(2-hydroxyethyl) terephthalate, were quantified using an analytical procedure with high-performance liquid chromatography (HPLC). The pH-dependent activities of Mipa-P, Kubu-P, and LCC were measured in the same mixture containing 100 mM Britton-Robinson universal pH buffer (instead of 50 mM glycine buffer) using relative absorbance at 270 nm.

### HPLC analytical procedure

Reaction samples were analyzed using a CBM-20A system (Shimadzu, Japan) equipped with a UV/Vis detector (SPD-20A) and a C18 column (Shimadzu Shim-pack GIST C18, 3 μm, 4.6 × 50 mm) protected by a guard cartridge (YMC ODS-A, 3 μm). Mobile phases A (0.1% formic acid) and B (100% methanol) were used with an isocratic flow of 33% concentration of B at a flow rate of 1 ml min<sup>-1</sup> and a column temperature of 40°C. The chromatograms were detected at 260 nm, and baseline was subtracted using arPLS (49) fit from Rampy (50). Subsequently, the peak areas were integrated. The amounts of the main products, terephthalic acid (Sigma-Aldrich, MO, USA, Cat. No.: 185361), mono(2-hydroxyethyl) terephthalic acid (Ambeed, IL, USA, Cat. No.: A875019), and bis(2-hydroxyethyl) terephthalate (Sigma-Aldrich, MO, USA, Cat. No.: 465151), were calculated using standard curves of the integrated area.

### Differential scanning fluorimetry

Fluorescent intensity for the enzyme melting curve was measured using the Protein Thermal shift dye kit (Applied Biosystems) in a Quantstudio 5 Real-time PCR instrument (Thermo Fisher Scientific). Each reaction mixture contained 5 μl of 1 mg ml<sup>-1</sup> enzyme solution, 100 mM Na<sub>2</sub>HPO<sub>4</sub>-HCl (pH 7.0), and 2.5 μl of 1× dye, and distilled water up to the final volume of 20 μl. Signal changes were monitored by increasing the temperature from 25° to 99°C at a rate of 0.1°C·s<sup>-1</sup>. The derivative or Boltzmann melting temperatures were calculated from the intensity change based on Boltzmann equation fitting or the peak of the first derivative curve. All data are shown in figs. S7, S15, and S19. Throughout the manuscript, we referred to the derivative melting temperature as T<sub>m</sub> when not specified by Boltzmann T<sub>m</sub>.

### Crystallization

Crystallization conditions of Mipa-P, Mipa-P<sup>M19</sup>, Kubu-P, Kubu-P<sup>P185V</sup>, and Kubu-P<sup>M12+P185V</sup> were screened using the sitting-drop vapor diffusion method. Each drop contained 1.0 μl of the

protein solution mixed with 1.0  $\mu\text{l}$  of reservoir solution, while the reservoir contained 50  $\mu\text{l}$  total volume. Initial screening was performed at 20°C using PEG/Ion HT and Index HT (Hampton Research), Wizard Classic I&II HT96 (Rigaku), Structure Screen 1&2 HT96 (Molecular Dimensions) and Wizard Cryo I&II HT96 (Rigaku). The crystals of Mipa-P and Mipa-P<sup>M19</sup> appeared under the following conditions: 20% polyethylene glycol (PEG) 8000 and 100 mM N-cyclohexyl-3-aminopropanesulfonic acid (CAPS)-NaOH (pH 10.5), and 20% PEG 3350 and 40 mM citric acid + 60mM BIS-Tris propane (pH 6.4), respectively. The crystals of Kubu-P and Kubu-P<sup>P185V</sup> appeared under the condition of 40% PEG 300, 100 mM imidazole-HCl (pH 8.0), and 200 mM zinc acetate. The crystals of Kubu-P<sup>M12+P185V</sup> appeared under the condition of 16% PEG 3350, 100 mM sodium acetate tribasic (pH 4.6), and 2% (v/v) Tacsimate (pH 4.0). The best quality crystals were soaked in reservoir solution containing 20% (v/v) glycerol for cryo-protection, then fished out with a loop and mounted on a goniometer at the beamline.

### Structure determination

Data were collected using beamline 7A of the Pohang Accelerator Laboratory (Republic of Korea) (57) with ADSC Q270 CCD or EIGER 9M detector at 100 K. The diffraction data were indexed, integrated, scaled, and processed using XDSGUI (52) and determination of space group with screw axis and resolution cutoff were performed using Aimless (53). The Mipa-P crystals belonged to space group  $P2_12_12_1$  with unit cell parameters of 49.07 Å, 103.41 Å, 131.88 Å, 90.0 Å, 90.0 β, and 90.0 γ. The highest resolution of the Mipa-P crystals obtained was 1.34 Å, and there are three molecules in the asymmetric unit. The Mipa-P<sup>M19</sup> crystals belonged to space group  $P12_11$  with unit cell parameters of 42.62 Å, 51.36 Å, 107.65 Å, 90.0 Å, 94.3 β, and 120.0 γ. The highest resolution of the Mipa-P<sup>M19</sup> crystals obtained was 1.89 Å, and there are two molecules in the asymmetric unit. The Kubu-P crystals belonged to space group  $I4_132$  with unit cell parameters of 159.1 Å, 159.10 Å, 159.10 Å, 90.0 Å, 90.0 β, and 90.0 γ. The highest resolution of the Kubu-P crystals obtained was 2.65 Å, and there is one molecule in the asymmetric unit. The Kubu-P<sup>P185V</sup> crystals belonged to space group  $I4_132$  with unit cell parameters of 157.22 Å, 157.22 Å, 157.22 Å, 90.0 Å, 90.0 β, and 90.0 γ. The highest resolution of the Kubu-P<sup>P185V</sup> crystals obtained was 1.7 Å, and there is one molecule in the asymmetric unit. The Kubu-P<sup>M12+P185V</sup> crystals belonged to space group  $I222$  with unit cell parameters of 64.96 Å, 81.65 Å, 83.73 Å, 90.0 Å, 90.0 β, and 90.0 γ. The highest resolution of the Kubu-P<sup>M12+P185V</sup> obtained was 1.15 Å, and there is one molecule in the asymmetric unit.

The phases of the diffraction patterns for the Mipa-P and Kubu-P crystals were determined

via molecular replacement using MOLREP (54) and PHASER (55) in CCP4 (56) and PHENIX (57), with the structure of CaPETase (PDB code: 7YM9) as the search model. For the variants (Mipa-P<sup>M19</sup>, Kubu-P<sup>P185V</sup>, and Kubu-P<sup>M12+P185V</sup>), the same methods were used employing their respective wild-type structures as search models. Structural models were built using WinCoot (58), and map refinements were performed using REFMAC5 (59). The statistics are summarized in table S2. Kubu-P<sup>M12</sup> structure was prepared using a homology model of Colabfold (60) from template structures of Kubu-P, Kubu-P<sup>P185V</sup>, and Kubu-P<sup>M12+P185V</sup>. The structures of Mipa-P, Mipa-P<sup>M19</sup>, Kubu-P, Kubu-P<sup>P185V</sup>, and Kubu-P<sup>M12+P185V</sup> were deposited in the Protein Data Bank, with PDB codes 8YTU, 8YTV, 8YTW, 8YTZ and 8YTY, respectively.

### Depolymerization via hydrolysis in bioreactor

The customized bioreactor (MARADO-PDA, BIOCONS Co. Ltd, South Korea) and thermostat bath circulator (RW3-1025, JEIO TECH, Daejeon, South Korea) (21) were used to evaluate the large-scale performance of enzymes in pH-stat hydrolytic PET depolymerization (table S6). The real-time values of pH during the reaction were monitored using a pH electrode (pH Sensor InPro3030/120, METTLER TOLEDO, Switzerland) and adjusted by titrating with NaOH solution using a PID controller to maintain pH 8.0 (figs. S11 and S23). For the pH optimization, we used conditions of 2% (w/w) PET and 3  $\text{mg}_{\text{enzyme}} \cdot \text{g}_{\text{PET}}^{-1}$  enzyme load. To account for stirring rate, alkalinity diffusion, and disodium terephthalate solubility, different titrant concentrations were used depending on the substrate load: 1 M for 2% load, 7.15 M for 20% load, and 4 M for 30% load. For the reactions under conditions of 2% (w/w) PET and 3  $\text{mg}_{\text{enzyme}} \cdot \text{g}_{\text{PET}}^{-1}$  enzyme load, 3.06 g PET powder was mixed with a 150 ml solution containing 100 mM  $\text{Na}_2\text{HPO}_4$  (pH 8.0) at a stirring rate of 300 rpm. Once the temperature reached the desired conditions, 1.836 ml of enzyme solution was added to the solution to initiate the reaction. For the reactions under harsh conditions of 20% (w/w) PET and 1  $\text{mg}_{\text{enzyme}} \cdot \text{g}_{\text{PET}}^{-1}$  enzyme load, 37.50 g PET powder and 7.5 ml of enzyme solution were added to the 150 ml solution with a stirring rate of 600 rpm. For the reactions under harsher conditions of 30% (w/w) PET and 0.58  $\text{mg}_{\text{enzyme}} \cdot \text{g}_{\text{PET}}^{-1}$  enzyme load, 64.29 g PET powder and 7.5 ml of enzyme solution were added to the 150 ml solution with a stirring rate of 600 rpm. Experimental data including temperature, pH change, titrant consumption, and revolutions per minute of the impeller, were recorded in real time on a personal computer. During the reactions, 500- $\mu\text{l}$  aliquots were sampled using a syringe, filtered through a PVDF syringe filter (0.22  $\mu\text{m}$ , SFPVDF013022CB, Futecs, South Korea), diluted to an optimal concentration, and analyzed using HPLC to quantify

the released monomers. The degree of depolymerization was calculated using two methods (37): the ester bond cleavage ratio (based on NaOH consumption) and the available monomer release ratio (based on NaOH consumption with the ratio of terephthalate to mono(2-hydroxyethyl) terephthalate). The weight loss values were determined by measuring the change in weight of the 0.22- $\mu\text{m}$  filter holder before and after filtration, with complete drying.

### Depolymerization via enzyme-catalyzed PET glycolysis

The all reactions for enzyme-catalyzed PET glycolysis were performed using a 99.5% ethylene glycol (EG) solution (Guaranteed Reagents grade, Daejung, South Korea). For the 1-day feasibility test, an enzyme-EG mixture containing 5  $\mu\text{l}$  of the 10.2  $\text{mg ml}^{-1}$  enzyme solution, 45  $\mu\text{l}$  of the lysis buffer, and 950  $\mu\text{l}$  of the EG solution was added to 50 mg of PET powder in a 1.5-ml microtube. After mixing for 10 s with a vortex mixer and centrifuging for 30 s at an RCF of 20,000, the reaction was initiated. The reaction in approximately 95% EG was conducted at 40°C without shaking. To evaluate the tolerance of Mipa-P<sup>M19</sup>, Kubu-P<sup>M12</sup>, LCC-ICCG to EG, the mixture was incubated for 0, 2, 4, 6, 8, and 10 days at temperatures ranging from 40° to 70°C. Then, the solution was loaded onto 50 mg of PET powder, and the reaction at 40°C was initiated in the same manner. For the buffer reagent test, we conducted enzyme-catalyzed PET glycolysis for 17 hours using a suspension containing 50 mg PET powder, 5  $\mu\text{l}$  enzyme solution, 950  $\mu\text{l}$  EG, and 45  $\mu\text{l}$  buffer solution (100 mM Tris or imidazole, dissolved in EG). In the ~99.5% condition, the relative activities of LCC-ICCG in enzyme-catalyzed PET glycolysis were significantly decreased compared to the experiments conducted in the ~95% EG condition (fig. S24). For the enzyme-catalyzed PET glycolysis reactions in the bioreactor at 40° and 50°C, 7.5 g PET powder, 0.81 g Tris, and 0.75 ml of the enzyme solution were added to the 150 ml solution with a stirring rate of 300 rpm, corresponding to a condition of 0.45% (w/w) PET 2  $\text{mg}_{\text{enzyme}} \cdot \text{g}_{\text{PET}}^{-1}$  enzyme load. It is worthy to note that EG is hygroscopic, but water vapor level was not tightly controlled during the reactions, especially for the bioreactor experiments. The degree of depolymerization was inferred from the concentration of produced BHET, using the mass per repeating unit of PET and the molecular weight of EG. The weight loss after the 100-hours reaction catalyzed by Kubu-P<sup>M12</sup> at 40°C was determined to be 17%, calculated based on the weight change of a 0.22- $\mu\text{m}$  filter holder before and after filtration with complete drying.

### REFERENCES AND NOTES

1. S. Yoshida et al., A bacterium that degrades and assimilates poly(ethylene terephthalate). *Science* **351**, 1196–1199 (2016). doi: 10.1126/science.aad6359; pmid: 26965627

2. V. Tournier *et al.*, An engineered PET depolymerase to break down and recycle plastic bottles. *Nature* **580**, 216–219 (2020). doi: [10.1038/s41586-020-2149-4](#); pmid: [32269349](#)
3. K. Oda, A. Wlodawer, Development of Enzyme-Based Approaches for Recycling PET on an Industrial Scale. *Biochemistry* **63**, 369–401 (2024). doi: [10.1021/acs.biochem.3c00554](#); pmid: [38285602](#)
4. M. Hunsen *et al.*, A Cutinase with Polyester Synthesis Activity. *Macromolecules* **40**, 148–150 (2007). doi: [10.1021/ma062095g](#)
5. R.-J. Müller, H. Schrader, J. Profe, K. Dresler, W.-D. Deckwer, Enzymatic Degradation of Poly(ethylene terephthalate): Rapid Hydrolyse using a Hydrolase from *T. fusca*. *Macromol. Rapid Commun.* **26**, 1400–1405 (2005). doi: [10.1002/marc.200500410](#)
6. A. Eberl *et al.*, Enzymatic surface hydrolysis of poly(ethylene terephthalate) and bis(benzoyloxyethyl) terephthalate by lipase and cutinase in the presence of surface active molecules. *J. Biotechnol.* **143**, 207–212 (2009). doi: [10.1016/j.biotech.2009.07.008](#); pmid: [19616594](#)
7. R. Wei *et al.*, Biocatalytic Degradation Efficiency of Postconsumer Polyethylene Terephthalate Packaging Determined by Their Polymer Microstructures. *Adv. Sci. (Weinh.)* **6**, 1900491 (2019). doi: [10.1002/adv.201900491](#); pmid: [31380212](#)
8. S. Sulaiman *et al.*, Isolation of a novel cutinase homolog with poly(ethylene terephthalate)-degrading activity from leaf-branch compost by using a metagenomic approach. *Appl. Environ. Microbiol.* **78**, 1556–1562 (2012). doi: [10.1128/AEM.06725-11](#); pmid: [22194294](#)
9. H. F. Son *et al.*, Rational Protein Engineering of Thermo-Stable PETase from *Ideonella sakaiensis* for Highly Efficient PET Degradation. *ACS Catal.* **9**, 3519–3526 (2019). doi: [10.1021/acscatal.9b00568](#)
10. B. C. Knott *et al.*, Characterization and engineering of a two-enzyme system for plastics depolymerization. *Proc. Natl. Acad. Sci. U.S.A.* **117**, 25476–25485 (2020). doi: [10.1073/pnas.2006753117](#); pmid: [32989159](#)
11. S. H. Lee *et al.*, Three-directional engineering of ISPETase with enhanced protein yield, activity, and durability. *J. Hazard. Mater.* **459**, 132297 (2023). doi: [10.1016/j.jhazmat.2023.132297](#); pmid: [37595467](#)
12. Y. Cui *et al.*, Computational Redesign of a PETase for Plastic Biodegradation under Ambient Condition by the GRAPE Strategy. *ACS Catal.* **11**, 1340–1350 (2021). doi: [10.1021/acscatal.0c05126](#)
13. H. Lu *et al.*, Machine learning-aided engineering of hydrolases for PET depolymerization. *Nature* **604**, 662–667 (2022). doi: [10.1038/s41586-022-04599-z](#); pmid: [35478237](#)
14. Y. Cui *et al.*, Computational redesign of a hydrolase for nearly complete PET depolymerization at industrially relevant high-solids loading. *Nat. Commun.* **15**, 1417 (2024). doi: [10.1038/s41467-024-45662-9](#); pmid: [38360963](#)
15. E. L. Bell *et al.*, Directed evolution of an efficient and thermostable PET depolymerase. *Nat. Catal.* **5**, 673–681 (2022). doi: [10.1038/s41929-022-00821-3](#)
16. S. Joo *et al.*, Structural insight into molecular mechanism of poly(ethylene terephthalate) degradation. *Nat. Commun.* **9**, 382 (2018). doi: [10.1038/s41467-018-02881-1](#); pmid: [29374183](#)
17. D. Danso *et al.*, New Insights into the Function and Global Distribution of Polyethylene Terephthalate (PET)-Degrading Bacteria and Enzymes in Marine and Terrestrial Metagenomes. *Appl. Environ. Microbiol.* **84**, e02773–e17 (2018). doi: [10.1128/AEM.02773-17](#); pmid: [29427431](#)
18. E. Erickson *et al.*, Sourcing thermotolerant poly(ethylene terephthalate) hydrolase scaffolds from natural diversity. *Nat. Commun.* **13**, 7850 (2022). doi: [10.1038/s41467-022-35237-x](#); pmid: [36543766](#)
19. Y. Joho *et al.*, Ancestral Sequence Reconstruction Identifies Structural Changes Underlying the Evolution of *Ideonella sakaiensis* PETase and Variants with Improved Stability and Activity. *Biochemistry* **62**, 437–450 (2023). doi: [10.1021/acs.biochem.2c00323](#); pmid: [35951410](#)
20. V. Vongsouthi *et al.*, Ancestral reconstruction of polyethylene terephthalate degrading 1 cutinases reveals a rugged and unexplored sequence-fitness landscape. *bioRxiv* 2024.04.25.591214 (2024). doi: [10.1101/2024.04.25.591214](#)
21. H. Hong *et al.*, Discovery and rational engineering of PET hydrolase with both mesophilic and thermophilic PET hydrolase properties. *Nat. Commun.* **14**, 4556 (2023). doi: [10.1038/s41467-023-40233-w](#); pmid: [37507390](#)
22. N. Lenfant *et al.*, ESTHER, the database of the  $\alpha/\beta$ -hydrolase fold superfamily of proteins: Tools to explore diversity of functions. *Nucleic Acids Res.* **41**, D423–D429 (2013). doi: [10.1093/nar/gks1154](#); pmid: [23193256](#)
23. J. N. Copp, E. Akiva, P. C. Babbitt, N. Tokuriki, Revealing Unexplored Sequence-Function Space Using Sequence Similarity Networks. *Biochemistry* **57**, 4651–4662 (2018). doi: [10.1021/acs.biochem.8b00473](#); pmid: [30052428](#)
24. R. Zallot, N. Oberg, J. A. Gerlt, The EFI Web Resource for Genomic Enzymology Tools: Leveraging Protein, Genome, and Metagenome Databases to Discover Novel Enzymes and Metabolic Pathways. *Biochemistry* **58**, 4169–4182 (2019). doi: [10.1021/acs.biochem.9b00735](#); pmid: [31553576](#)
25. W. Yeung *et al.*, Tree visualizations of protein sequence embedding space enable improved functional clustering of diverse protein superfamilies. *Brief. Bioinform.* **24**, bbac619 (2023). doi: [10.1093/bib/bbac619](#); pmid: [36642409](#)
26. W. J. Wilbur, D. J. Lipman, Rapid similarity searches of nucleic acid and protein data banks. *Proc. Natl. Acad. Sci. U.S.A.* **80**, 726–730 (1983). doi: [10.1073/pnas.80.3.726](#); pmid: [6572363](#)
27. P. Shannon *et al.*, Cytoscape: A software environment for integrated models of biomolecular interaction networks. *Genome Res.* **13**, 2498–2504 (2003). doi: [10.1101/gr.1239303](#); pmid: [14597658](#)
28. C. Roth *et al.*, Structural and functional studies on a thermostable poly(ethylene terephthalate) degrading hydrolase from *Thermobifida fusca*. *Appl. Microbiol. Biotechnol.* **98**, 7815–7823 (2014). doi: [10.1007/s00253-014-5672-0](#); pmid: [24728714](#)
29. J. Arnlung Bååth *et al.*, Structure-function analysis of two closely related cutinases from *Thermobifida cellulosilytica*. *Biotechnol. Bioeng.* **119**, 470–481 (2022). doi: [10.1002/bit.27984](#); pmid: [34755331](#)
30. H.-Y. Sagong *et al.*, Implications for the PET decomposition mechanism through similarity and dissimilarity between PETases from *Rhizobacter gummiphilus* and *Ideonella sakaiensis*. *J. Hazard. Mater.* **416**, 126075 (2021). doi: [10.1016/j.jhazmat.2021.126075](#); pmid: [34492896](#)
31. X. Xi *et al.*, Secretory expression in *Bacillus subtilis* and biochemical characterization of a highly thermostable poly(ethylene terephthalate) hydrolase from bacterium HR29. *Enzyme Microb. Technol.* **143**, 109715 (2021). doi: [10.1016/j.enzmictec.2020.109715](#); pmid: [33375975](#)
32. X. Hu, U. Thumarat, X. Zhang, M. Tang, F. Kawai, Diversity of polyester-degrading bacteria in compost and molecular analysis of a thermoactive esterase from *Thermobifida alba* AHK119. *Appl. Microbiol. Biotechnol.* **87**, 771–779 (2010). doi: [10.1007/s00253-010-2555-x](#); pmid: [20393707](#)
33. C. M. Carr *et al.*, Purification and biochemical characterization of SM14est, a PET-hydrolyzing enzyme from the marine sponge-derived *Streptomyces* sp. SM14. *Front. Microbiol.* **14**, 1170880 (2023). doi: [10.3389/fmicb.2023.1170880](#); pmid: [37250061](#)
34. C. Thawai, S. Tanasupawat, T. Kudo, *Micromonospora pattalpongensis* sp. nov., isolated from a Thai mangrove forest. *Int. J. Syst. Evol. Microbiol.* **58**, 1516–1521 (2008). doi: [10.1099/ij.s.0.65410-0](#); pmid: [18599686](#)
35. C. Suriyachadkun, W. Ngaemthao, S. Chunhametha, T. Tamura, J.-J. Sanglier, *Kutzneria buriramensis* sp. nov., isolated from soil, and emended description of the genus *Kutzneria*. *Int. J. Syst. Evol. Microbiol.* **63**, 47–52 (2013). doi: [10.1099/ij.s.0.036533-0](#); pmid: [22328609](#)
36. X. Han *et al.*, Structural insight into catalytic mechanism of PET hydrolase. *Nat. Commun.* **8**, 2106 (2017). doi: [10.1038/s41467-017-02255-z](#); pmid: [29235460](#)
37. G. Arnal *et al.*, Assessment of Four Engineered PET Degrading Enzymes Considering Large-Scale Industrial Applications. *ACS Catal.* **13**, 13156–13166 (2023). doi: [10.1021/acscatal.3c02922](#); pmid: [37881793](#)
38. A. M. Castro, A. Carniel, A novel process for poly(ethylene terephthalate) depolymerization via enzyme-catalyzed glycolysis. *Biochem. Eng. J.* **124**, 64–68 (2017). doi: [10.1016/j.bej.2017.04.011](#)
39. H. Yao *et al.*, Solubility determination and thermodynamic modeling of bis-2-hydroxyethyl terephthalate (BHET) in different solvents. *Chin. J. Chem. Eng.* **45**, 294–300 (2022). doi: [10.1016/j.cjche.2021.03.024](#)
40. T. Oeser *et al.*, High level expression of a hydrophobic poly(ethylene terephthalate)-hydrolyzing carboxylesterase from *Thermobifida fusca* KW3 in *Escherichia coli* BL21(DE3). *J. Biotechnol.* **146**, 100–104 (2010). doi: [10.1016/j.jbiotec.2010.02.006](#); pmid: [20156495](#)
41. D. Ribitsch *et al.*, Hydrolysis of polyethyleneterephthalate by p-nitrobenzylesterase from *Bacillus subtilis*. *Biotechnol. Prog.* **27**, 951–960 (2011). doi: [10.1002/btpr.610](#); pmid: [21574267](#)
42. T. B. Thomsen, C. J. Hunt, A. S. Meyer, Influence of substrate crystallinity and glass transition temperature on enzymatic degradation of poly(ethylene terephthalate) (PET). *N. Biotechnol.* **69**, 28–35 (2022). doi: [10.1016/j.nbt.2022.02.006](#); pmid: [35247624](#)
43. P. Yuan, B. Liu, H. Sun, Optimization of the Crystallization Process of Bis(2-hydroxyethyl) Terephthalate. *Cryst. Res. Technol.* **56**, 2100025 (2021). doi: [10.1002/crat.202100025](#)
44. F. Sievers, D. G. Higgins, Clustal Omega for making accurate alignments of many protein sequences. *Protein Sci.* **27**, 135–145 (2018). doi: [10.1002/pro.3290](#); pmid: [28884485](#)
45. D. Maglott, J. Ostell, K. D. Pruitt, T. Tatusova, Entrez Gene: Gene-centered information at NCBI. *Nucleic Acids Res.* **39**, D52–D57 (2011). doi: [10.1093/nar/gkr937](#); pmid: [21115458](#)
46. O. Bányi *et al.*, Catalogue of Life Checklist (Version 2024-03-25), Catalogue of Life (2024); <https://www.catalogueoflife.org/data/download>
47. M. Gostev *et al.*, The BioSample Database (BioSD) at the European Bioinformatics Institute. *Nucleic Acids Res.* **40**, D64–D70 (2012). doi: [10.1093/nar/gkr937](#); pmid: [22096232](#)
48. J. J. Almagro Armenteros *et al.*, SignalP 5.0 improves signal peptide predictions using deep neural networks. *Nat. Biotechnol.* **37**, 420–423 (2019). doi: [10.1038/s41587-019-0036-z](#); pmid: [30778233](#)
49. S.-J. Baek, A. Park, Y.-J. Ahn, J. Choo, Baseline correction using asymmetrically reweighted penalized least squares smoothing. *Analyst* **140**, 250–257 (2015). doi: [10.1039/C4AN01061B](#); pmid: [25382860](#)
50. C. Le Losq, Rampy: a Python library for processing spectroscopic (IR, Raman, XAS...) data. Version 1, Zenodo (2018); <https://doi.org/10.5281/zenodo.1168730>
51. S.-Y. Park, S.-C. Ha, Y.-G. Kim, The Protein Crystallography Beamlines at the Pohang Light Source II. *Biodesign* **5**, 30–34 (2017).
52. W. Brehm, J. Triviño, J. M. Krahn, I. Usón, K. Diederichs, XDSGUI: A graphical user interface for XDS, SHEXL and ARCIMBOLDO. *J. Appl. Cryst.* **56**, 1585–1594 (2023). doi: [10.1107/S1600576723007057](#); pmid: [37791359](#)
53. P. R. Evans, G. N. Murshudov, How good are my data and what is the resolution? *Acta Crystallogr. D* **69**, 1204–1214 (2013). doi: [10.1107/S0907444913000061](#); pmid: [23793146](#)
54. A. Vagin, A. Teplyakov, MOLREP: An Automated Program for Molecular Replacement. *J. Appl. Cryst.* **30**, 1022–1025 (1997). doi: [10.1107/S0021889897006766](#)
55. A. J. McCoy, Solving structures of protein complexes by molecular replacement with Phaser. *Acta Crystallogr. D* **63**, 32–41 (2007). doi: [10.1107/S0907444906045975](#); pmid: [17164524](#)
56. J. Aguirre *et al.*, The CCP4 suite: Integrative software for macromolecular crystallography. *Acta Crystallogr. D* **79**, 449–461 (2023). doi: [10.1107/S2059798323003595](#); pmid: [37259835](#)
57. P. D. Adams *et al.*, PHENIX: A comprehensive Python-based system for macromolecular structure solution. *Acta Crystallogr. D* **66**, 213–221 (2010). doi: [10.1107/S0907444909052925](#); pmid: [20124702](#)
58. P. Emsley, K. Cowtan, Coot: Model-building tools for molecular graphics. *Acta Crystallogr. D* **60**, 2126–2132 (2004). doi: [10.1107/S0907444904019158](#); pmid: [15572765](#)
59. G. N. Murshudov *et al.*, REFMAC5 for the refinement of macromolecular crystal structures. *Acta Crystallogr. D* **67**, 355–367 (2010). doi: [10.1107/S0907444911001314](#)
60. M. Mirdita *et al.*, ColabFold: Making protein folding accessible to all. *Nat. Methods* **19**, 679–682 (2022). doi: [10.1038/s41592-022-01488-1](#); pmid: [35637307](#)
61. philosopi-everthine, philosopi-everthine/philosopi-public: philosopi-public v1.0.0-alpha (PHILOSOPHI), Zenodo (2024); <https://doi.org/10.5281/zenodo.12574368>

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CJ CheilJedang. The other authors declare that they have no competing interests. **Data and materials availability:** All data are available in the main text or the supplementary materials. Information on the newly determined structures can be found at the Protein Data Bank with accession codes 8YTU, 8YTV, 8YTW, 8YTZ, and 8YTY. The python code for the neighborhood analysis module for landscaping is deposited at Zenodo (61). **License information:** Copyright © 2025 the authors, some rights reserved; exclusive licensee American Association for the Advancement of Science. No claim to original US government works. <https://www.science.org/about/science-licenses-journal-article-reuse>

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## Landscape profiling of PET depolymerases using a natural sequence cluster framework

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### Editor's summary

Polyethylene terephthalate (PET), which is used in drinking bottles, fibers, and many other applications, is one of a few plastics that can be broken down to its constituent monomers by naturally occurring enzymes. Seo *et al.* developed a landscape profiling method to identify and characterize the potential of microbial enzymes to degrade these plastics. Screening of nearly 2000 candidate enzymes across several families yielded a view of landscape proficiency and stability, from which they could identify peaks for further testing and optimization. Two enzymes were engineered with sequential mutagenesis and exhibited excellent performance relative to benchmarks, especially under the harsh conditions that are ideal for use in recycling applications. —Michael A. Funk and Iyinoluwa Sofowora

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